



13.11.4 Wrap-Up: Ticket Out the Door

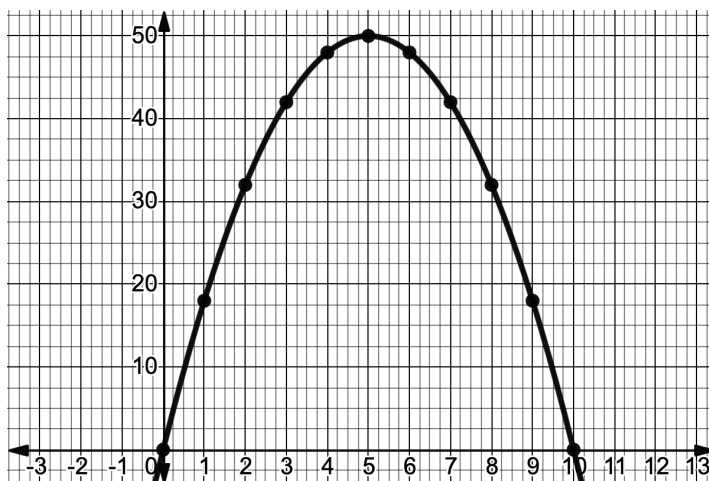
A rain gutter is made from a 20 inch (in.) aluminum sheet with the edges folded up, as shown.

Flat Aluminum Sheet	Sheet Folded for Rain Gutter
 20 in.	 (20-2x) in.

The function $A(x) = x(20 - 2x)$ models the cross-sectional

area of a rain gutter made from the sheet, where x is the height of the rain gutter.

The graph of the function is shown.



1. What is the average rate of change for the interval of heights of 1 in. to 5 in.?
2. Interpret the average rate of change for the interval of 1 in. to 5 in. in terms of the context.



Unit 13, Lesson H2: Combining Functions



Benchmark

MA.912.F.3.1 Given a mathematical or real-world context, combine two functions, limited to linear and quadratic, using arithmetic operations. When appropriate, include domain restrictions for the new function.



Learning Targets

- ☐ I can combine two functions using arithmetic operations.
- ☐ I can determine domain restrictions for a real-world context where two functions are combined.



13.H2.1 Warm-Up: Book Sales

1. The number of books sold, in millions, in the U.S. and the population of the U.S., in millions, from 2010 to 2016 are shown in the table.

Years Since 2010	Number of Books Sold in the U.S. (millions)	Population of the U.S. (millions)
0	2,530	309.35
1	2,400	311.64
2	2,730	313.99
3	2,720	316.23
4	2,700	318.62
5	2,710	321.04
6	2,700	323.41

- a. What do you notice?
- b. How many books were sold per person in 2010 and 2016?
- c. What do these values tell you about book sales?



13.H2.2 Collaboration: Real-World Functions

1. Penelope sets up an Etsy store to sell necklaces. Each listing costs \$0.20, with 5% of sales being paid to Etsy. Penelope opens a shop with x listings. The supplies for making each necklace are \$3.85 per listing. Penelope paid \$43.58 for the tools she will use to make the necklaces. Penelope sells each necklace for \$15. The function $C(x) = 0.2x + 3.85x + 0.05(15x) + 43.58$ represents Penelope's costs. The function $R(x) = 15x$ represents Penelope's revenue.

- a. What is a reasonable domain for $C(x)$?
- b. What is a reasonable domain for $R(x)$?

Profit is the amount of revenue minus the cost.

- c. Write a function, $P(x)$, that can be used to determine Penelope's profit.
- d. What is a reasonable domain for $P(x)$?

2. Tim and Jessica plan to rent a camper van for a vacation. The van costs \$183 per day. The rental includes 120 miles for free and then charges \$0.39 per mile. The function $V(m) = 0.39(m - 120) + 183$ models the cost of the van rental, V , in terms of miles traveled, m . Researching the cost of gasoline for the van, they determine gas will cost \$1.32 for each mile they travel. The company charges a \$45 fee for the first tank of gas. The function $G(m) = 1.32m + 45$ models the cost of gasoline for their vacation.

- What is a reasonable domain for $V(m)$?
- What is a reasonable domain for $G(m)$?
- Determine the function that gives the total cost, $C(m)$, in dollars, based on the miles traveled.
- What is a reasonable domain for $C(m)$?

3. A clothing store is stocking up before holiday sales. Their wholesaler will sell the first 120 clothing items for \$9 each, and then charge \$0.12 less for each set of 20 items over 120. The function $N(x) = 120 + x$ represents the total number of clothing items purchased, where x represents the number of clothing items over 120 items. The cost of each clothing item is $C(x) = 9 - 0.12\left(\frac{x}{20}\right)$.

- Write a new function that represents the total cost of clothing items the store purchases.
- Discuss with your partner what constraints should be on the value of x . Then, write the constraints.
- Ask two classmates the constraints they determined. Record their constraints and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have your classmate initial next to your summary stating that your summary was correct.

Constraint	My Summary of Their Reason	Initials

- Discuss with your partner the constraints given by your classmates. If necessary, revise the constraints you wrote.





13.H2.3 Exploration: Combining Functions

Tables of values for the functions $f(x) = 3x^2 - 4x - 2$ and $g(x) = 1 - 3x$ are shown.

$f(x) = 3x^2 - 4x - 2$		$g(x) = 1 - 3x$	
x	y	x	y
-2	18	-2	7
-1	5	-1	4
0	-2	0	1
1	-3	1	-2
2	2	2	-5
3	13	3	-8

1. Complete the table for $y = 3x^2 - 4x - 2 + 1 - 3x$.

x	-2	-1	0	1	2	3
y						

2. What strategy did you use to complete the table?

3. Find $f(x) + g(x)$. Write your answer in standard form.

4. Discuss with your partner how adding the y -values in the tables of $f(x)$ and $g(x)$ connect with the y -values for $3x^2 - 4x - 2 + 1 - 3x$. Summarize your discussion.

5. Complete the table for $f(x) - g(x)$ using the tables of $f(x)$ and $g(x)$.

x	-2	-1	0	1	2	3
y						

6. Find $f(x) - g(x)$. Write your answer in standard form.

7. Find $f(x) \cdot g(x)$. Write your answer in standard form.

8. Complete the table for $f(x) \cdot g(x)$.

x	-2	-1	0	1	2	3
y						

9. Find $\frac{f(x)}{g(x)}$.

10. What value of x will result in a denominator of 0?

11. Discuss with your partner why the denominator cannot equal 0. Summarize your discussion.

12. The value of x that makes the denominator equal to 0 is a restriction on the domain. Complete the restriction.

$$x \neq \underline{\hspace{2cm}}$$

13. Complete the table for $\frac{f(x)}{g(x)}$.

x	-2	-1	0	1	2	3
y						



13.H2.4 Bring it Together: Function Operations

Combining functions using function notation is similar to operations with polynomials. When combining functions from a real-world context, it is important to consider the domain.

- Addition of functions $f(x) + g(x) = (f + g)(x)$
- Subtraction of functions $f(x) - g(x) = (f - g)(x)$
- Multiplication of functions $f(x) \cdot g(x) = (f \cdot g)(x)$
- Division of functions $\frac{f(x)}{g(x)} = \left(\frac{f}{g}\right)(x)$

1. Use the functions $h(y) = 11y - 5y^2 + y$ and $m(y) = 2y - 8$ to determine each new function. Write your answer in standard form.

$$w(y) = (h + m)(y)$$

$$k(y) = (h - m)(y)$$

$$b(y) = (h \cdot m)(y)$$

$$c(y) = \left(\frac{h}{m}\right)(y)$$

2. Write the restriction on the domain for $c(y)$.





13.H2.5 Wrap-Up: Ticket Out the Door

1. A company's mathematician determines the function $P(x) = 750 - 25x$ can be used to determine the price of a seminar on data management, where x is the number of times the price decreases \$25. The company knows 1,000 people will attend. For each decrease in price, 80 more people will be able to afford the seminar. The function $N(x) = 1000 + 80x$ models the number of people who can attend the seminar, where x is the number of times the price decreases \$25.
 - a. Find $R(x)$, where $R(x) = P(x) \cdot N(x)$. Write the answer in standard form.
 - b. What are the constraints for the function $R(x)$?

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